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**UNITED STATES INTERNATIONAL TRADE COMMISSION
WASHINGTON, D.C.**

**Before the Honorable Clark S. Cheney
Chief Administrative Law Judge**

In the Matter of

**CERTAIN SEMICONDUCTOR
DEVICES, PRODUCTS CONTAINING
SAME, AND COMPONENTS THEREOF**

Investigation No. 337-TA-1500

RESPONDENTS' OPENING BRIEF ON CLAIM CONSTRUCTION

PUBLIC VERSION
TABLE OF CONTENTS

	Page
I. INTRODUCTION	1
II. LEGAL BACKGROUND	2
III. CLAIM CONSTRUCTION.....	3
A. “layer” / “first liner layer” / “refractory metal layer”	4
B. “intermix region”	11
C. “from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide”	14
D. “adjacent to” and “the channel is displaced from [a/the] first heavily doped region adjacent to the first edge of the gate”	20
E. “the drift isolation region is disposed under the gate”	26
F. “the channel having a channel length that is controlled by ion implantation”	30
G. “edge of the gate”	35
IV. CONCLUSION.....	39

PUBLIC VERSION
TABLE OF AUTHORITIES

	Page(s)
Cases	
<i>Berkheimer v. HP Inc.</i> , 881 F.3d 1360 (Fed. Cir. 2018)	22
<i>Certain Cellular Base Station Commc’n Equip., Components Thereof, & Prods. Containing Same</i> , Inv. No. 337-TA-1397, Order No. 16, 2024 WL 5245312 (Nov. 5, 2024) (Moore, ALJ).....	3
<i>Deere & Co. v. Bush Hog, LLC</i> , 703 F.3d 1349 (Fed. Cir. Dec. 4, 2012).....	29, 34
<i>Dow Chem. Co. v. Nova Chems. Corp.</i> , 803 F.3d 620 (Fed. Cir. 2015)	15, 19
<i>Eidos Display, LLC v. AU Optronics Corp.</i> , 779 F.3d 1360 (Fed. Cir. 2015)	3
<i>Icon Health & Fitness, Inc. v. Polar Electro Oy</i> , 656 F. App'x. 1008 (Fed. Cir. 2016).....	24
<i>Infinity Comp. Prods., Inc. v. Oki Data Americas, Inc.</i> , 987 F.3d 1053 (Fed. Cir. 2021)	21
<i>Innova/Pure Water, Inc. v. Safari Water Filtration Sys., Inc.</i> , 381 F.3d 1111 (Fed. Cir. 2004)	2
<i>Interval Licensing LLC v. AOL, Inc.</i> , 766 F.3d 1364 (Fed. Cir. 2014)	3, 14, 22
<i>Markman v. Westview Instruments, Inc.</i> , 52 F.3d 967 (Fed. Cir. 1995)	3
<i>Nautilus, Inc. v. Biosig Instruments, Inc.</i> , 572 U.S. 898 (2014)	1, 3, 19
<i>Phillips v. AWH Corp.</i> , 415 F.3d 1303 (Fed. Cir. 2005) (<i>en banc</i>).....	2, 3
<i>Teva Pharms. USA, Inc. v. Sandoz, Inc.</i> , 789 F.3d 1335 (Fed. Cir. 2015)	3, 15, 19
<i>TVnGO Ltd. (BVI) v. LG Elecs. Inc.</i> , 861 F. App'x. 453 (Fed. Cir. 2021)	21
<i>Vitronics Corp. v. Conceptronic, Inc.</i> , 90 F.3d 1576 (Fed. Cir. 1996)	2

PUBLIC VERSION
EXHIBIT LIST

Exhibit	Description
A	Declaration of Dr. Sanjay K. Banerjee Regarding Claim Construction
B	Declaration of Dr. Frank Ernst Regarding Claim Construction
C	Declaration of Dr. David K. Choi Regarding Claim Construction
D	Dr. Sanjay K. Banerjee Curriculum Vitae
E	Dr. Frank Ernst Curriculum Vitae
F	Dr. David K. Choi Curriculum Vitae
1	Complainant's Infringement Contentions, Ex. A4
2	<i>Refractory Metal</i> , ScienceDirect, https://www.sciencedirect.com/topics/materials-science/refractory-metal (last visited August 24, 2026)
3	<i>Titanium</i> , Royal Society of Chemistry, https://periodic-table.rsc.org/element/22/titanium (last visited August 24, 2026)
4	U.S. Patent Application Publication No. 2012/0161324 A1 ("Heinrich")
5	U.S. Patent Application Publication No. 2014/0027883 A1 ("Hu")
6	U.S. Patent Application Publication No. 2015/0221740 A1 ("Alptekin")
7	<i>Intellectual Ventures I LLC, v. Canon Inc.</i> , C.A. No. 11-792-SLR, Memo. Ord. re Claim Constr. (D. Del. Apr. 10, 2014), Dkt. No. 252
8	<i>Bridgelux, Inc., v. Cree, Inc.</i> , C.A. No. 06-6495-PJH, Ord. Construing Claims (N.D. Cal. Aug. 15, 2008), Dkt. No. 109
9	<i>Advanced Tech. Incubator, Inc. v. Sharp Corporation</i> , C.A. No. 2:7-cv-468, Claim Constr. Ord. (E.D. Tex. Mar. 11, 2009), Dkt. No. 117
12	Merriam-Webster's Collegiate Dictionary (11th Ed.), 2003 (definition of "intermix")
13	<i>Excerpt</i> of Prosecution History of the U.S. Patent No. 9,865,546 (GF-ITC-00004771 – GF-ITC-00005023)
14	<i>Excerpt</i> of Prosecution History of the U.S. Patent No. 10,707,167 (GF-ITC-00004128 – GF-ITC-00004309)
15	Chow, et al., <i>Silicide Formation from Laser Thermal Processing of Ti/Co Bilayers</i> (2004)
16	Email between counsel for Complainant and Respondents re claim construction
17	Praneet Adusumilli et al, <i>Silicide-phase evolution and platinum redistribution during silicidation of Ni_{0.95}Pt_{0.05}/Si(100) specimens</i> , 112 J. Appl. Phys. 064307 (2012) ("Adusumilli")
18	'235 patent <i>Excerpt</i> from Complainant GlobalFoundries U.S. Inc.'s Second Supplemental Objections and Responses to Respondents' First Set of Interrogatories (Nos. 1-73) ("GF validity contentions")
19	Complainant's Infringement Contentions, Ex. A1
21	U.S. Patent Application Publication No. 2008/0191291A1 ("Wu '291") (TOWER 1500 00034458)
22	U.S. Patent Application Publication No. 2009/0020811A1 ("Voldman '811") (TOWER 1500 00034410)
23	This Exhibit Was Intentionally Left Blank

PUBLIC VERSION
EXHIBIT LIST (cont.)

24	'983 patent prosecution history: July 16, 2012 Application (GF-ITC-00004497 – 4571)
25	U.S. Patent No. 6,979,864 (“Negoro”)
26	U.S. Patent No. 7,935,992 (“Chang”)
27	'983 patent prosecution history: Jan. 25, 2013 Amendment (GF-ITC-00004596 – 4610)
28	'983 patent prosecution history: October 26, 2012 Non-Final Rejection (GF-ITC-00004576 – 4593)
29	U.S. Patent No. 8,053,319 (“Liu”)
30	U.S. Patent No. 8,222,130 (“Zhang”)
31	<i>Excerpt</i> from <i>Micro-Electronic Circuits</i> , Adel S. Sedra et al., CBS College Publishing (1982) (TOWER_1500_00040579)
32	'983 patent <i>Excerpt</i> from Complainant GlobalFoundries U.S. Inc.’s Second Supplemental Objections and Responses to Respondents’ First Set of Interrogatories (Nos. 1-73) (“GF validity contentions”)
33	<i>Excerpt</i> from <i>Micro-Electronic Circuits</i> , Adel S. Sedra et al., Saunders College Publishing (1991)
34	Ashwin Shah et al., <i>A 4Mb DRAM with cross point trench transistor cell</i> , SC-21:5 IEEE J. ISSCC Dig. Tech. Papers, 618-626 (1986)
35	Shaw Banerjee & D. Mark Bordelon, <i>A Model for the Trench Transistor</i> , ED-34:12 IEEE Transactions on Electron Devices 2485, 2485-2492 (1987)
36	Shaw Banerjee et al, <i>Leakage Mechanisms in the Trench Transistor DRAM Cell</i> , ED-35:1 IEEE Transactions on Electron Devices, 108-116 (1988)

I. INTRODUCTION

Complainant GlobalFoundries U.S. Inc. (“Complainant” or “GlobalFoundries”) asserts infringement of U.S. Patent Nos. 8,330,235 (“the ’235 patent”); 8,507,983 (“the ’983 patent”); 9,093,425 (“the ’425 patent”); 9,865,546 (“the ’546 patent”); 10,062,748 (“the ’748 patent”); and 10,707,167 (“the ’167 patent”). The parties dispute the claim constructions and/or indefiniteness of certain terms in the asserted claims of the ’546, ’167, ’235, and ’983 patents.¹ Respondents Tower Semiconductor Ltd., Tower Partners Semiconductor, Ltd., Tower Semiconductor Italy S.r.l., Tower US Holdings, Inc., Tower Semiconductor San Antonio, Inc., Tower Semiconductor NPB Holdings, Inc., Tower Semiconductor Newport Beach, Inc., and Newport Fab LLC (collectively, “Respondents”) assert that all claims of the ’546, ’167, ’235, and ’983 patents are invalid for indefiniteness. These asserted claims fail to define the scope of the alleged inventions with reasonable certainty to a person of ordinary skill in the art (“POSA”) under the Supreme Court’s guidance in *Nautilus, Inc. v. Biosig Instruments, Inc.*, 572 U.S. 898 (2014). If not held to be indefinite, the CALJ should adopt Respondents’ proposed constructions. The table below summarizes Respondents’ indefiniteness and claim construction positions.²

Claim Term	Associated Patent Claims	Proposed Construction
“layer” / “first liner layer” / “refractory metal layer”	’546 patent, claims 1-14 ’167 patent, claims 1, 3, and 4	Indefinite or, if not indefinite: “a separate and defined [first liner / refractory metal] coating of material”

¹ Complainant asserts infringement of claims 1, 5-12, and 15-19 of the ’235 patent; claims 1-7, 13-18, and 20 of the ’983 patent; claims 1-9 of the ’425 patent; claims 1-14 of the ’546 patent; claims 1-12, 14, and 16-18 of the ’748 patent; and claims 1-6 of the ’167 patent.

² Pursuant to Ground Rule 7.1, Respondents narrowed their list of disputed terms to the eight terms addressed herein. Respondents reserve the right to argue indefiniteness for all other terms identified as being indefinite in Respondents’ Invalidity Contentions. Respondents also reserve the right to propose constructions for additional terms or different constructions for the terms addressed herein based on pending and forthcoming disputes regarding invalidity, infringement, or any other issue that requires claim construction.

Claim Term	Associated Patent Claims	Proposed Construction
“intermix region”	’546 patent, claims 1, 2, 6, 8, 9, and 13 ’167 patent, claims 1, 2, and 6	Indefinite
“from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide”	’235 patent, claims 1 and 10	Indefinite
“adjacent to”	’983 patent, claims 1, 5, 13, 17, and 20	Indefinite
“the channel is displaced from [a/the] first heavily doped region adjacent to the first edge of the gate”	’983 patent, claims 1, 13, and 20	Indefinite
“the drift isolation region is disposed under the gate”	’983 patent, claims 4 and 16	“the drift isolation region is positioned at least in part under the gate”
“the channel having a channel length that is controlled by ion implantation”	’983 patent, claims 7 and 18	“the channel having a channel length that is formed by implanting ions into the substrate to form a channel well”
“edge of the gate”	’983 patent, claims 1, 2, 6, 13, 14, 15, and 20	“edge of the gate electrode over the gate dielectric”

Complainant proposes plain-and-ordinary meaning constructions for all disputed terms.

II. LEGAL BACKGROUND

The Federal Circuit has instructed courts to first consider intrinsic evidence (the patent and its file history) when construing claims. “It is a ‘bedrock principle’ of patent law that ‘the claims of a patent define the invention to which the patentee is entitled the right to exclude.’” *Phillips v. AWH Corp.*, 415 F.3d 1303, 1312 (Fed. Cir. 2005) (quoting *Innova/Pure Water, Inc. v. Safari Water Filtration Sys., Inc.*, 381 F.3d 1111, 1115 (Fed. Cir. 2004)). In addition, “the specification ‘is always highly relevant to the claim construction analysis. Usually, it is dispositive; it is the single best guide to the meaning of a disputed term.’” *Id.* at 1315 (quoting *Vitronics Corp. v. Conceptronic, Inc.*, 90 F.3d 1576, 1582 (Fed. Cir. 1996)). “In addition to consulting the

specification, . . . a court ‘should also consider the patent’s prosecution history, if it is in evidence.’” *Phillips*, 415 F.3d at 1317 (quoting *Markman v. Westview Instruments, Inc.*, 52 F.3d 967, 980 (Fed. Cir. 1995), *aff’d*, 517 U.S. 370 (1996)). Extrinsic evidence, including dictionaries, expert testimony, and prior art references, may be considered when meaning is not clear from the intrinsic evidence. *Phillips*, 415 F.3d at 1317–1318; *Eidos Display, LLC v. AU Optronics Corp.*, 779 F.3d 1360, 1365 (Fed. Cir. 2015).

“[A] patent is invalid for indefiniteness if its claims, read in light of the specification delineating the patent, and the prosecution history, fail to inform, with *reasonable certainty*, those skilled in the art about the scope of the invention.” *Teva Pharms. USA, Inc. v. Sandoz, Inc.*, 789 F.3d 1335, 1340-41 (Fed. Cir. 2015) (“*Teva*”) (quoting *Nautilus*, 572 U.S. at 901). “A claim is required to provide objective boundaries for those of skill in the art, and a claim term may be indefinite if it might mean several different things and no informed and confident choice is among the contending definitions.” *Certain Cellular Base Station Commc'n Equip., Components Thereof, & Prods. Containing Same*, Inv. No. 337-TA-1397, Ord. No. 16, 2024 WL 5245312, at *4 (Nov. 5, 2024) (quoting *Interval Licensing LLC v. AOL, Inc.*, 766 F.3d 1364, 1371 (Fed. Cir. 2014) (citation modified)).

III. CLAIM CONSTRUCTION

The parties have agreed that the term “outermost edges . . . vertically coincident to each other” in claim 2 of the ’425 patent means “outermost edges . . . vertically aligned with each other.” The parties dispute the meanings of the following terms.

A. “layer” / “first liner layer” / “refractory metal layer”

Disputed Claim Term	Complainant’s Proposed Construction	Respondents’ Proposed Construction
“layer” / “first liner layer” / “refractory metal layer” (’546 patent, claims 1-14; ’167 patent, claims 1, 3, and 4)	Plain and ordinary meaning; not indefinite.	Indefinite or, if not indefinite: “a separate and defined [first liner / refractory metal] coating of material”

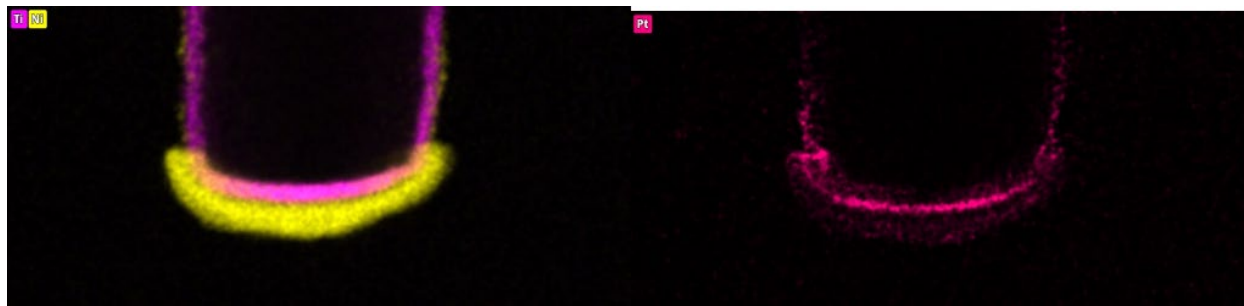
These terms have plain-and-ordinary meanings, as reflected in Respondents’ alternative proposed construction. *See* Banerjee Decl., ¶¶ 52-53. Complainant apparently disagrees with Respondents’ alternative constructions but refuses to explain what the plain-and-ordinary meanings of these terms are. Instead, as shown by Complainant’s Infringement and Domestic Industry Contentions, Complainant appears interested in using these terms in ways that would render them indefinite for failure to provide reasonable certainty regarding the scope of the ’546 and ’167 patents’ claims. *See id.*, ¶¶ 53-67.

The terms “layer,” “first liner layer,” and “refractory metal layer” have plain-and-ordinary meanings: “a separate and defined [first liner / refractory metal] coating of material.” *Id.*, ¶¶ 52-53. These meanings are supported by dictionary definitions. For example, one dictionary defines “layer” to mean “[a] single thickness, coating, or stratum spread out or covering a surface.” Ex. 7, *Intellectual Ventures I LLC, v. Canon Inc.*, C.A. No. 11-792-SLR, Memo. Ord. re Claim Constr. at 9-10 (D. Del. Apr. 10, 2014), Dkt. No. 252 (citing Webster’s Dictionary). Another dictionary defines “layer” as “[a] defined thickness which is part of a material, or which surrounds it.” Ex. 8, *Bridgelux, Inc., v. Cree, Inc.*, C.A. No. 06-6495-PJH, Ord. Construing Claims at 10 (N.D. Cal. Aug. 15, 2008), Dkt. No. 109 (citing Wiley Electrical and Electronics Engineering Dictionary (John Wiley & Sons, 2004)).

One court considering similar indefiniteness concerns observed: “if a ‘layer’ were not

considered ‘separate and distinct,’ then the Parties could argue at trial that the term ‘layer’ be interpreted as several layers which are co-mingled and which are not considered either separate or distinct. Such a construction would contradict the ordinary meaning of the term ‘layer.’” Ex. 9, *Advanced Tech. Incubator, Inc. v. Sharp Corp.*, C.A. No. 2:7-cv-00468, Claim Constr. Ord. at 17 (E.D. Tex. Mar. 11, 2009), Dkt. No. 117. Similarly, here, there is no way by which a POSA would be able to understand the ’546 or ’167 patents’ “layer” terms if they are not interpreted to mean “a separate and defined [first liner / refractory metal] coating of material.” Banerjee Decl., ¶¶ 52, 54.

Complainant’s Infringement and Domestic Industry Contentions demonstrate the indefiniteness of these “layer” terms if they are not interpreted in accordance with their plain-and-ordinary meanings as proposed by Respondents. *Id.*, ¶¶ 54-58. For example, [REDACTED] uses nickel, platinum, and titanium at different stages of fabrication. *See* Ex. 1, Complainant’s Infringement Contentions, Ex. A4 at 4-7. Rather than identifying a single one of these materials as the “first liner layer,” Complainant’s Infringement Contentions appear to arbitrarily identify **all three** for the “first liner layer” claim element, as shown in the unidentified and unexplained screenshot below from Complainant’s Contentions (“Ti” representing alleged titanium, “Ni” representing alleged nickel, and “Pt” representing alleged platinum). *Id.* 5.



Complainant apparently has no idea which of these, if any, could be a “first liner layer” as used in the ’546 and ’167 patents. Banerjee Decl., ¶ 54. Complainant appears to want to take advantage of the patents’ failure to delineate what constitutes a layer (versus more than one layer) to contort

the claim language to allegedly read on Respondents’ accused processes. But Complainant’s apparent interpretations of these terms make it impossible to determine which coating(s) should be considered a “first liner layer” when there is not a one-to-one mapping of different types of materials to the individual elements recited in the claims. *Id.*, ¶¶ 54-55.

Complainant also appears interested in eliminating any meaning for the word “layer” by arguing that a single coating could be multiple layers. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

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[REDACTED]

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[REDACTED]

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[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] These sorts of words games demonstrate the meaninglessness and indefiniteness of the “layer” terms as construed by Complainant. Banerjee Decl., ¶ 58. A POSA would have no idea how to assess infringement or non-infringement in view of the arbitrary nature of line drawing enabled by Complainant’s interpretations of these terms. *Id.*

Complainant seems to rely on a lack of clarity in the intrinsic record to interpret the “layer” terms indefinitely. Banerjee Decl., ¶¶ 59-61. For example, claim 1 of the ’546 patent requires “a portion of the first liner layer to diffuse into the upper portion of the semiconductor substrate to form a first intermix region at the upper portion of the semiconductor substrate.” *See* ’546 patent, claim 1; *see also id.* claim 8 (reciting similar “first intermix region” over source and drain regions); Banerjee Decl., ¶ 61. Under Complainant’s interpretation, a POSA would have been unable to assess infringement of this claim element if “first liner layer” were not interpreted to mean “a separate and defined [first liner] coating of material,” in other words, the first coating of material that is deposited to “conformally coat the contact opening.” Banerjee Decl., ¶ 61. Arbitrarily

combining the first coating of material with *other, subsequent* coatings of materials (as suggested in Complainant’s Infringement and Domestic Industry Contentions) would make it impossible for the POSA to assess what needs to be diffused into the upper portion of the semiconductor substrate to form a “first intermix region,” as claimed. *See* ’546 patent, claim 1; Banerjee Decl., ¶¶ 55, 61. In addition, if materials *other than* the very first material deposited over a contact opening could be included in the alleged “first liner layer,” then it would be impossible for the POSA to determine what materials must “conformally coat the contact opening” as claimed. *See* ’546 patent, claim 1; Banerjee Decl., ¶¶ 55, 61.

The claims of the ’546 and ’167 patents require the claimed first liner layers and refractory metal layers to coat other components, such as the contact opening (in the case of the first liner layer) or the first liner layer or intermix region (in the case of the refractory metal layer). *See* ’546 patent, claims 1, 8; ’167 patent, claim 1. To assess infringement, therefore, the claimed layers must be interpreted to be *separate* layers to enable a POSA to understand the relative positions and orientations of the layers as compared to the claims. *See* Ex. 9, *Advanced Tech.*, C.A. No. 2:7-cv-00468, Claim Constr. Ord. at 18-20 (“[W]hile these layers are defined in terms of their functions (*e.g.*, ‘conductive layer’), the inventor used the terms ‘atop’ and ‘upon’ to define the placement of the layers. The plain meaning of the terms ‘atop’ and ‘upon’ indicate that this process involves placing one layer on another layer and that although there may not be a ‘precise line of demarcation,’ the layers are considered separate layers. . . . This is especially true where, as here, the drawings include solid lines indicating separate layers[.]”); Banerjee Decl., ¶¶ 52-67.

Complainant appears to arbitrarily define “layer” to include any combination and/or division of separate coatings of material(s). But if Complainant’s arbitrary view is adopted, a POSA would have been unable to distinguish between a “first liner layer” and “refractory metal

layer” in the case where more than two (for the ’546 patent) or three (for the ’167 patent) coatings of materials could qualify either alone or in combination to practice either or both of these limitations. Banerjee Decl., ¶ 56. To be definite, these terms must be construed to mean “a separate and defined [first liner / refractory metal] coating of material.” *Id.*, ¶ 52. Any other interpretation would render the claim language meaningless, as demonstrated by Complainant’s Contentions.

The ’546 and ’167 patents fail to provide any guidance on what a “layer” is beyond this plain-and-ordinary meaning. For example, the independent claims of the ’546 and ’167 patents refer to a “dielectric layer,” “first liner layer,” and “refractory metal layer.” *See* ’546 patent, claims 1, 8; ’167 patent, claim 1. However, the patents provide no guidance as to how to identify any layer or how to distinguish a given layer from any other component or structure recited in the claims. Banerjee Decl., ¶¶ 59-60. The specification compounds this uncertainty by noting that the “dielectric layer” could be a “single layer,” two layers, or “more than two” layers, without any description of when a POSA should consider two or more layers to be separate layers versus a single combined layer. ’546 patent, 4:56-62; Banerjee Decl., ¶ 60. But if this type of generic description were used, a POSA could not reliably identify any “layer.” For instance, if Respondents’ position is not adopted, layers within a larger layer could comprise the same material, making it difficult, if not impossible, for a POSA to distinguish between layers when attempting to assess infringement or non-infringement. Banerjee Decl., ¶ 60. Similarly, a POSA would not know whether or when to combine different, distinguishable layers of material to consider them to be a single “layer.” *Id.*

Complainant’s undisclosed understanding of “layer” to be something other than a separate and defined . . . coating of material” also renders indefinite the larger “first liner layer” and

“refractory metal layer” phrases. The terms “first liner layer” and “refractory metal layer” are recited in all claims of the ’546 and ’167 patents without any guidance in the patents as to what these terms mean or how to distinguish between them. Banerjee Decl., ¶¶ 61-66. For example, the claims provide no guidance as to how to identify any first liner or refractory metal layer or how to distinguish any of these layers from other component recited in the claims. *Id.* If these “layers” are not “separate and defined . . . coatings of material,” a POSA would not know whether (a) a single coating of material is one layer or two (or more) layers, (b) two separate coatings of the same material are one layer or two (or more) layers, (c) two separate coatings of different materials are one layer or two (or more) layers, etc. *Id.*

The patents’ common specification also fails to provide adequate guidance to depart from Respondents’ plain-and-ordinary meanings. For example, regarding the “first liner layer,” the common specification includes a general description that the “first liner layer . . . conformally coat[s] each contact opening” and that it may include various materials, but fails to clarify whether the “first liner layer” is a portion of a coating, one single coating, or two or more separate coatings. *See, e.g.,* ’167 patent, 5:58-65; Banerjee Decl., ¶¶ 62-63. Complainant apparently seeks to exploit the specification’s lack of guidance to argue that “layer” could mean “any number of layers.” Banerjee Decl., ¶ 66. Once again, however, “if a ‘layer’ were not considered ‘separate and distinct,’ then the Parties could argue at trial that the term ‘layer’ be interpreted as several layers which are co-mingled and which are not considered either separate or distinct. Such a construction would contradict the ordinary meaning of the term ‘layer.’” Ex. 9, *Advanced Tech. Incubator*, C.A. No. 2:7-cv-00468 at 17.

In sum, if not interpreted in accordance with Respondents’ plain-and-ordinary-meaning construction, the “layer” terms are indefinite.

B. “intermix region”

Disputed Claim Term	Complainant’s Proposed Construction	Respondents’ Proposed Construction
“intermix region” (’546 patent, claims 1, 2, 6, 8, 9, and 13; ’167 patent, claims 1, 2, and 6)	Plain and ordinary meaning; not indefinite.	Indefinite

The asserted claims of the ’546 and ’167 patents are indefinite because the term “intermix region” fails to inform a POSA about the scope of the alleged inventions with reasonable certainty, when read in light of the patents’ specifications and prosecution histories. Banerjee Decl., ¶ 68.

A POSA would not have recognized the term “intermix region” as having any particular meaning in the art. *Id.*, ¶ 69. Thus, a POSA would have looked to the common specification of the ’546 and ’167 patents to attempt to determine the meaning of “intermix region.” *Id.* The specification, however, provides no guidance as to when two (or more) layers are sufficiently “intermixed” to qualify as an “intermix region.” *Id.*, ¶¶ 69-76.

First, the patents lack sufficient guidance as to what it means to be “intermixed.” *Id.*, ¶ 79. For example, the figure below from Ex. 15, Chow, et al., *Silicide Formation from Laser Thermal Processing of Ti/Co Bilayers* (2004), at 2 (Fig. 3), shows layers of TiO_x, a “bilayered structure” of “amorphous CoTi silicide” “on a highly textured CoTi silicide layer, and Si substrate.”

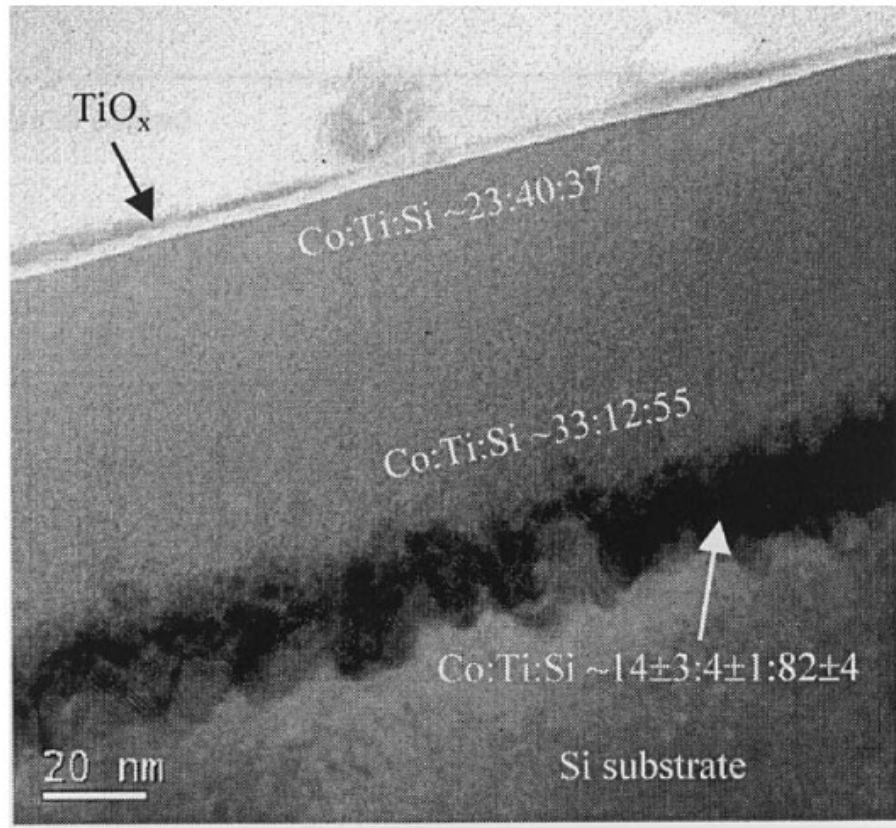


Figure 3. XTEM of the Ti/Co/Si sample after pulsed laser irradiation at 0.6 J/cm² shown in Fig. 1. A bilayered structure, *i.e.*, an amorphous CoTi silicide layer on a highly textured CoTi silicide layer, can be seen.

The interfaces between the various layers shown above are uneven. Banerjee Decl., ¶ 79. Do these uneven surfaces form part of an “intermix region” as claimed? *Id.* If so, which portions of which layers should be included in the “intermix region”? *Id.* If these uneven surfaces are not sufficient to form an “intermix region,” what more is required? *Id.* For example, is it enough that atoms from one coating are present in another coating, or must there be more meaningful interaction between the atoms from two layers. *Id.* The patents lack adequate guidance on this issue. *Id.*

Second, even to the extent the patents provide guidance on what it means to be “intermixed” (they do not), the patents provide no guidance on the *amount* of “intermixing” that

is required. *Id.*, ¶ 80. For example, assuming that “intermixing” includes diffusion of atoms from one coating into another coating, is there some threshold level of diffusion that is required? *Id.* The patents include a statement that “very little if any metal diffusion is needed” to form the intermix region, but lack any guidance on what qualifies as “very little.” ’546 patent, 6:39-43; Banerjee Decl., ¶ 80. For example, if a single atom from one layer diffuses into another layer, is this sufficient to qualify as an “intermix region”? Banerjee Decl., ¶ 80. If a region is 98% of one material, and only 2% of a second material, is this enough to comprise an “intermix region”? *Id.* One portion of the common specification notes that the “intermix region” may have around 50% mixing between two materials. ’546 patent, 6:19-21; Banerjee Decl., ¶ 80. But the specification goes on to identify numerous factors that may impact the amount of intermixing. ’546 patent, 6:21-28 (“type of material being deposited, the composition of source 124a and drain 124b, the energy of the atoms of the material being deposited, and/or the conditions of the deposition technique (temperature, pressure, reaction, atmosphere, etc.)”), 6:28-31 (“etch selectivity”), 6:31-34 (“amorphous” versus “crystalline”); Banerjee Decl., ¶ 80. While the common specification identifies factors that may impact the amount of intermixing, it puts no meaningful limits on what amount of mixing would or would not qualify as creating an “intermix region.” Banerjee Decl., ¶ 80.

Third, the patents do not explain if any kind of purposeful intermixing of materials is required or whether incidental intermixing (which may not be intended or desired as part of manufacturing) could create an intermix region. *Id.*, ¶ 81.

The patents’ lack of guidance on these issues only becomes more complex when more than two materials are required to be part of an “intermix region.” *Id.*, ¶ 70. For example, the “intermix region” in the ’167 patent’s claims and the “second intermix region” in claims 6 and 13 of the ’546

patent must be composed of materials from a “first liner layer,” “semiconductor substrate,” and “refractory metal layer.” See ’167 patent, claim 1; ’546 patent, claims 6, 13; Banerjee Decl., ¶ 70. But there is no explanation in the patents as to whether this second intermix region may have, for example, intermixing of the refractory metal layer with the first liner layer at the top of the alleged intermix region (with no intermixed material from the semiconductor substrate included) and intermixing of the first liner layer with material from the semiconductor substrate at the bottom of the alleged intermix region (with no intermixed material from the refractory metal layer included), or, alternatively, whether substantially uniform intermixing of all three materials is required throughout, or, alternatively, whether portions of relatively lower (or no) intermixing may be included. Banerjee Decl., ¶ 70.

Because the phrase “intermix region” fails to provide objective boundaries for those of skill in the art, the claims must be held invalid as indefinite. *Interval Licensing*, 766 F.3d at 1371. Banerjee Decl., ¶ 68.

C. “from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide”

Disputed Claim Term	Complainant’s Proposed Construction	Respondents’ Proposed Construction
“from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide” (’235 patent, claims 1 and 10)	Plain and ordinary meaning; not indefinite.	Indefinite

Each of the claims in the ’235 patent require determining the amount of platinum (specifically, between about 8-12 wt. %) at a specific location: the “nickel silicide surface at a silicon interface.” The issue here is that there are multiple methodologies for determining the location of this surface/interface, where different methodologies will lead to different results.

Because the specification and prosecution history are silent regarding which methodology applies to the '235 patent claims, the location of the “nickel silicide surface at a silicon interface” cannot be determined with sufficient certainty, and the claims must be found indefinite. *Dow Chem. Co. v. Nova Chems. Corp.*, 803 F.3d 620, 634–35 (Fed. Cir. 2015) (“*Dow*”) (holding that the existence of multiple methods for assessing a claim limitation, where the multiple methods leads to different results, without guidance in the patent or the prosecution history as to which method should be used renders the claims indefinite); *Teva*, 789 F.3d at 1343–45 (finding that the term “molecular weight” rendered claims indefinite because molecular weight could be ascertained by three possible measures and the claims did not indicate which measure to use).

While the terms “nickel silicide surface” and “silicon interface” are not necessarily unfamiliar to a POSA, a POSA would recognize that there are multiple different ways of identifying the location of each surface/interface. Ernst Decl., ¶¶ 30-60. For example, the paper “Silicide-phase evolution and platinum redistribution during silicidation of Ni_{0.95}Pt_{0.05}/Si(100) specimens” by Adusumilli et. al (“Adusumilli”) uses atomic probe tomography (APT) and uses two different concentrations of silicon in atomic percent to as reference points to identify the “nickel-silicide/Si(100) interface”: (1) 74 at. % and (2) 75 at. %. Ex. 17, Adusumilli at 5, Ernst Decl., ¶¶ 51-60. Adusumilli then chooses a point representing the silicide/Si interface as shown by the vertical dotted line in the images below. *Id.*

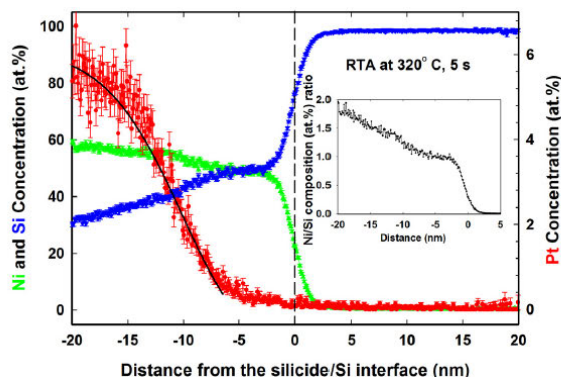


FIG. 4. Proximity histogram displaying the concentrations of Ni (green, ▲), Si (blue, ▼), and Pt (red, •) versus film depth for a 10 nm thick $\text{Ni}_{0.95}\text{Pt}_{0.05}/\text{Si}(100)$ film subjected to RTA at 320°C for 5 s. Concentrations were calculated for a series of 75 at. % Si isoconcentration surfaces moving into the thin film. On the abscissa zero represents the nickel-silicide/Si(100) heterophase interface. The black solid-line is a half-Gaussian fit to the Pt concentration profile. The inset diagram shows the Ni-to-Si concentration ratio (atomic percent) versus film depth.

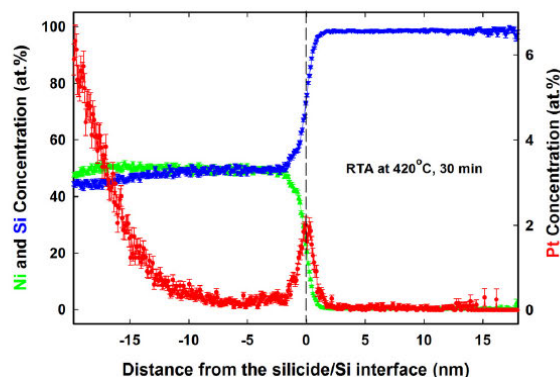


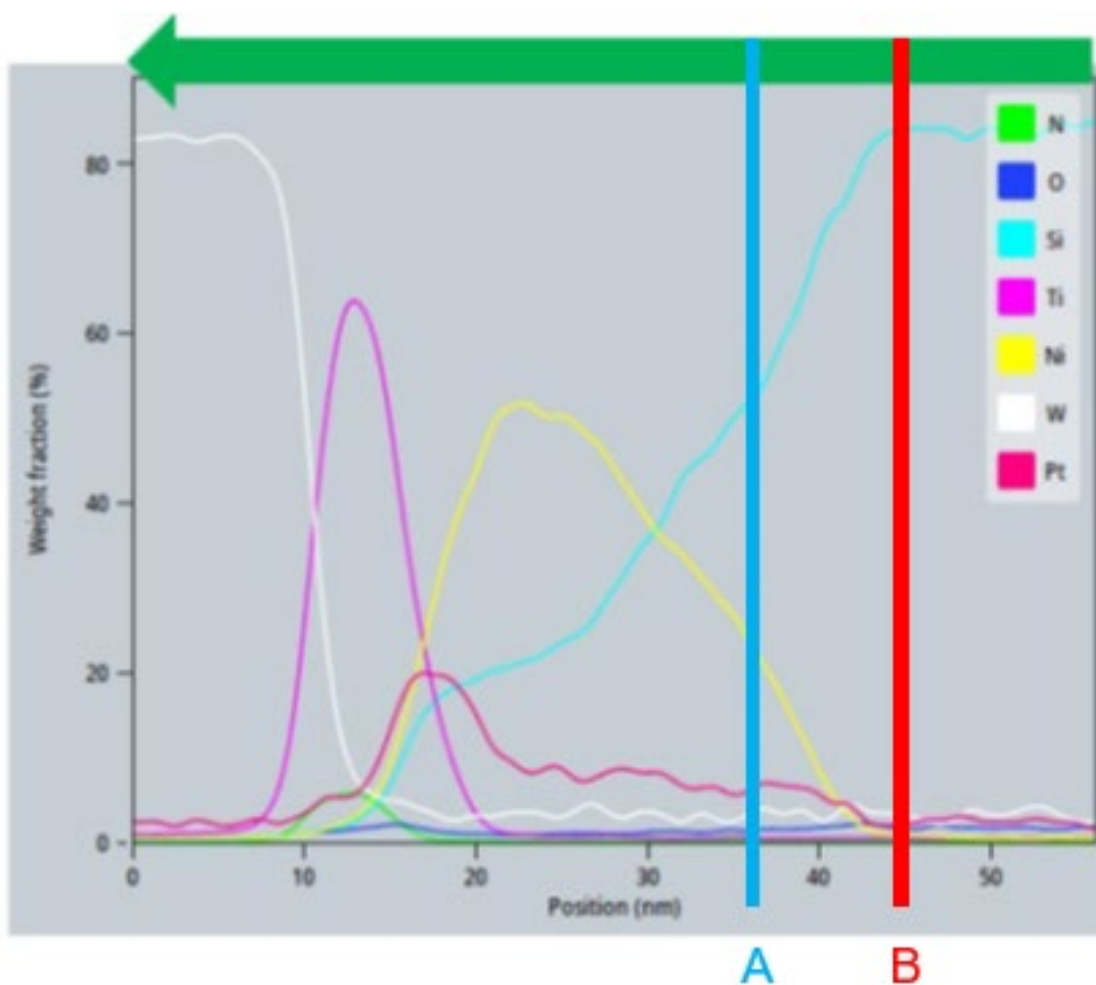
FIG. 6. Proximity histogram displaying the concentrations of Ni (green, ▲), Si (blue, ▼), and Pt (red, •) versus film depth for a 10 nm $\text{Ni}_{0.95}\text{Pt}_{0.05}/\text{Si}(100)$ thin film subjected to RTA at 420°C for 30 min. Concentrations were calculated for a series of 74 at. % Si isoconcentration surfaces moving into the thin film. Zero on the abscissa represents the nickel-silicide/Si(100) interface.

Ex. 17, Adusumilli at Figs. 4, 6. As an alternative method, a POSA could perform analysis of electron refraction patterns to identify changes in crystal structure, and use the changes in crystal structure to identify an approximate location of the desired surface/interface. Ernst Decl., ¶¶ 51-60. Using either of these exemplary methodologies (the Adusumilli method or the electron refraction pattern method) would yield different results. *Id.* This issue is further compounded by the multiple ways to actually take measurements at specific locations, which causes further ambiguity and potential for differences in results. *Id.*, ¶¶ 36-40.

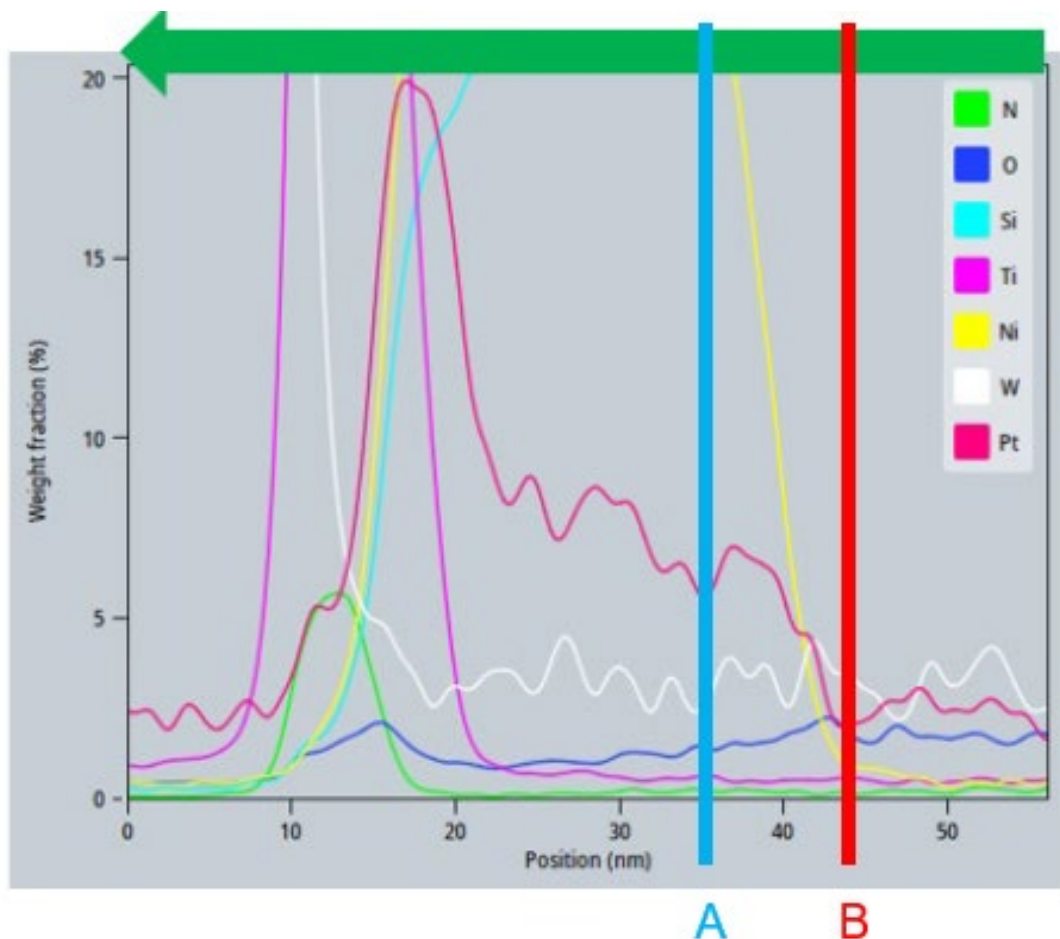
The claims, specification, and file history provide no guidance on what methodology should be used to locate/measure the claimed surface/interface. For example, the term “silicon interface” is used a single time in the specification and the term “interface” is used twice: (1) “The resulting NiSi layer 401 has a composition gradient of platinum from about 8 wt. % to about 12 wt. %, such as about 10 wt. %, at the **silicon interface** 403 to about 30 wt. % to about 70 wt. %, such as about 30 wt. % to about 50 wt. %, at the opposite surface” and (2) “The two-step anneal therefore reduces the time at which annealing occurs at a high temperature, thereby reducing the rough **interfaces** between the NiSi and the underlying Si layer.” ’235 patent, 7:4-12, 7:33-46

(emphasis added). Neither portion of the specification explains how to determine the location of the “silicon interface” or provides guidance on what methodology should be used for the claims.³

The problem with the '235 patent's lack of guidance is that differing methodologies for determining the location of the “silicon interface” will lead to different results. Ernst Decl., ¶¶ 51-60. Complainant's Infringement Contentions are illustrative. *Id.* In its Infringement Contentions, Complainant pointed vaguely to the figure copied below in support of its argument that there is about 8-12 at % of platinum at the “silicon interface”:



³ The same problem exists for the phrase “opposite surface,” which is only used twice in the specification. '235 patent, 7:4-12, 8:41-52.



Ex. 19, Complainant's Infringement Contentions, Ex. A1 at 9-10 (Annotated); *see also* Ernst Decl., ¶¶ 51-60.⁴ But depending on what methodology is used to define the location of the "silicon interface," the result changes. For example, if the "silicon interface" is located at either Line A or Line B, Complainant's graph would show the product being analyzed is outside the scope of the claims: Line A corresponds to approximately 7 weight % platinum and Line B corresponds to 3 weight % platinum, both of which are below the 8-12% platinum required by the claims. *Id.* Accordingly, Complainant is apparently using some undisclosed, unspecified test for determining

⁴ Complainant has failed to produce the underlying reverse engineering documents or provide any details on how these tests were performed. Respondents do not adopt any of the values listed in the following paragraph and disagree that these figures are representative of the amount of any element in Respondents' products. The discussion is solely for purposes of showing the indefiniteness of the asserted claims of the '235 patent.

the location of the silicon interface.⁵ *Id.* But the arbitrary nature of Complainant’s selection only further reaffirms the indefinite nature of the claims. *Id.*

The lack of guidance in the specification and prosecution history on how to locate the “silicon interface” renders the claims indefinite as a matter of law. The facts here are analogous to *Teva* and *Dow*. In *Teva*, the Federal Circuit held that the term “molecular weight” rendered the claims indefinite because (i) there were three possible ways measuring molecular weight: M_p , M_n , and M_w , and (ii) the claims, specification, and file history did not indicate which measure to use. *Teva*, 789 F.3d at 1343–45. Similarly, in *Dow* the Federal Circuit found the claims were invalid because (i) there were multiple methodologies for determining the “slope of strain hardening” that was necessary for the infringement analysis, with different methodologies leading to different results, and (ii) neither the patent nor the prosecution history provided guidance as to which method should be used. *Dow*, 803 F.3d 634–35. Like in *Teva* and *Dow*, here (i) there are multiple methodologies for determining the location of the “silicon interface,” with different methodologies leading to different results, and (ii) the intrinsic evidence does not provide any explanation on what methodology should be used. Because the identification of the “silicon interface” is determinative of infringement, a POSA could obtain materially different results depending on what location is used and how it is measured. Such a “zone of uncertainty” regarding the scope of the claims, and regarding whether a product does or does not infringe, renders the term indefinite. *Nautilus*, 572 U.S. at 909 (2014).

⁵ During the meet-and-confer process, Respondents’ counsel asked for clarity regarding *where* the silicon interface was located. Complainant’s counsel refused to provide any guidance outside of repeatedly stating that the term should be given its plain-and-ordinary meaning. But Complainant’s counsel could not explain what that plain-and-ordinary meaning was and refused to identify where in the specification Respondents’ counsel could look to understand the scope of the term “silicon interface.” The parties’ communications after the meet-and-confer process further demonstrate this. *See* Ex. 16.

D. “adjacent to” and “the channel is displaced from [a/the] first heavily doped region adjacent to the first edge of the gate”

Disputed Claim Term	Complainant’s Proposed Construction	Respondents’ Proposed Construction
“adjacent to” (’983 patent, claims 1, 5, 13, 17, and 20)	Plain and ordinary meaning; not indefinite.	Indefinite
“the channel is displaced from [a/the] first heavily doped region adjacent to the first edge of the gate” (’983 patent, claims 1, 13, and 20)	Plain and ordinary meaning; not indefinite.	Indefinite

The term “adjacent to” in the asserted claims of the ’983 patent is indefinite because a POSITA would have lacked reasonable certainty about the range of spatial relationships covered by its scope. A POSITA would have had no objective boundary as to when elements are “adjacent to” each other and when elements are not “adjacent to” each other. The claims recite elements that are “adjacent to” each other across different distances and despite the presence of other elements in between. Furthermore, the claims recite that the “first heavily doped region” and the “first edge of the gate” must satisfy the contradictory requirements of being both “adjacent to” and “displaced from” each other. When Tower raised these issues with Complainant during the meet-and-confer process, Complainant failed to articulate (or even attempt to articulate) the scope of the “adjacent to” claim language. The Court should find all asserted claims of the ’983 patent indefinite based on the “adjacent to” claim language.

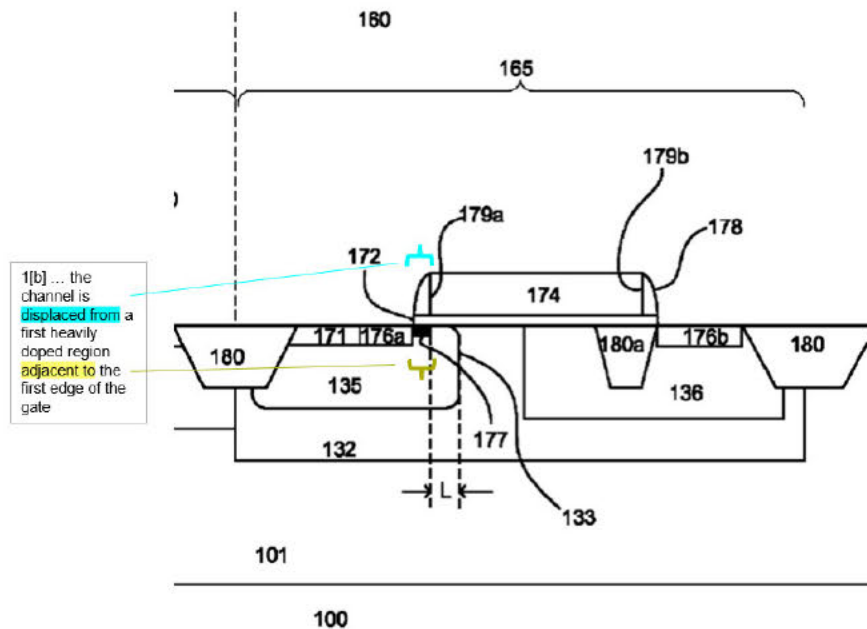
Nothing in the claims provides guidance for the scope of “adjacent to.” The asserted claims of the ’983 patent simply recite various elements that are “adjacent to” each other. Choi Decl., ¶¶ 42-45. Claims 1, 13, and 20 (all independent claims) recite “a doped channel well disposed in the substrate adjacent to a first edge of the gate.” ’983 patent, 13:20-21, 14:8-9, 14:47-48. Claims

1, 13, and 20 recite “a first heavily doped region adjacent to the first edge of the gate.” *Id.*, 13:25-26, 14:14-16, 14:52-54. Claims 1, 13, and 20 also recite “a doped drift well adjacent to a second edge of the gate.” *Id.*, 13:27, 14:17, 14:56. Claims 5 and 17 additionally recite “the first and a second heavily doped regions are provided adjacent to the gate.” *Id.*, 13:37-39, 14:28-30.

Rather than providing guidance on the scope of “adjacent to,” the claims only obfuscate that scope by reciting that certain elements (the “first heavily doped region” and the “first edge of the gate”) are both “adjacent to” and “displaced from” each other at the same time. *TVnGO Ltd. (BVI) v. LG Elecs. Inc.*, 861 F. App’x. 453, 459–60 (Fed. Cir. 2021) (affirming indefiniteness based on “intra-patent inconsistency”); *Infinity Comp. Prods., Inc. v. Oki Data Americas, Inc.*, 987 F.3d 1053, 1060 (Fed. Cir. 2021) (affirming indefiniteness given “flat contradiction” in the intrinsic record). Claims 1, 13, and 20 (all independent claims) require that the “first heavily doped region” be “adjacent to” the “first edge of the gate.” ’983 patent, 13:25-26, 14:14-16, 14:52-54. Further, claims 1, 13, and 20 require that the “first heavily doped region” be “displaced from” the “channel.” *Id.* Claims 1, 13, and 20 also require that “the first edge of the gate and channel edge beneath the gate define a channel of the device.” *Id.*, 13:23-24, 14:12-14, 14:51-53. In other words, the “first heavily doped region” must be both (1) “adjacent to” the first edge of the gate and (2) “displaced from” the first edge of the gate (as one of the points that “define[s]” the channel). Choi Decl., ¶¶ 48-49; *see also id.*, ¶¶ 29-37. A POSITA would have lacked reasonable certainty for how elements could be both “adjacent to” and “displaced from” each other. *Id.*, ¶¶ 46-49.

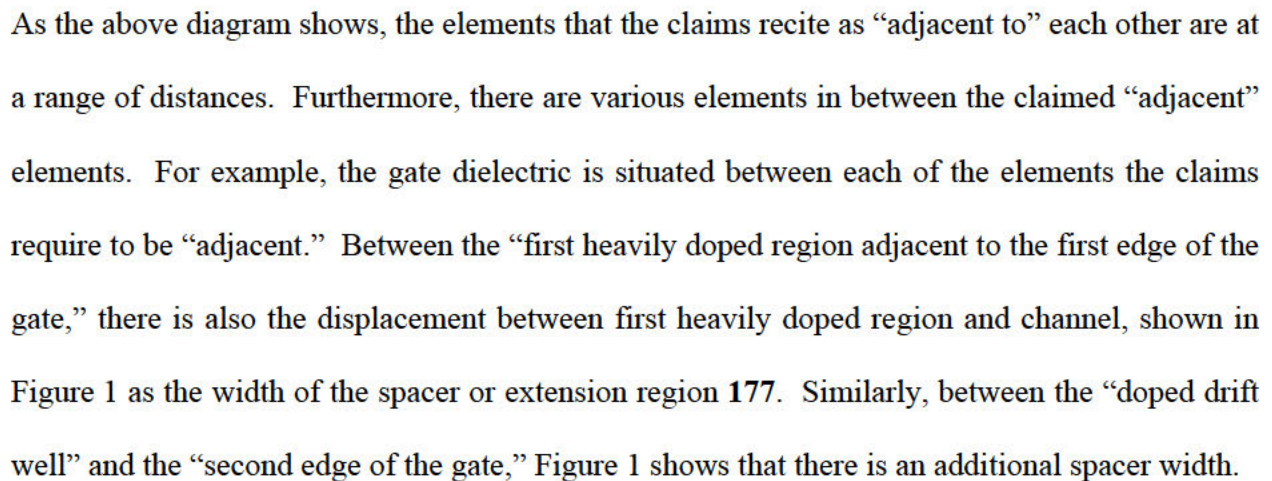
'983 patent

Fig. 1 (annotations added) (Choi Decl., ¶ 47)



The specification provides no objective guidance for either the terms “adjacent to” or “displaced from.” *Interval Licensing*, 766 F.3d at 1374 (“The specification offers no indication, thus leaving the skilled artisan to consult the unpredictable vagaries of any one person’s opinion” (citation modified).); *Berkheimer v. HP Inc.*, 881 F.3d 1360, 1364 (Fed. Cir. 2018) (affirming indefiniteness given “no point of comparison for skilled artisans to determine an objective boundary”). The specification merely repeats the claims’ recitation that certain elements are “adjacent to” other elements. The specification provides no elaboration or guidance on the meaning or scope of the “adjacent to” claim language. Further, the language “displaced from” or “displace” never appears in the specification—instead, only appearing (without explanation or guidance) in the claims.

The diagrams in the specification only underline the lack of reasonable certainty in the

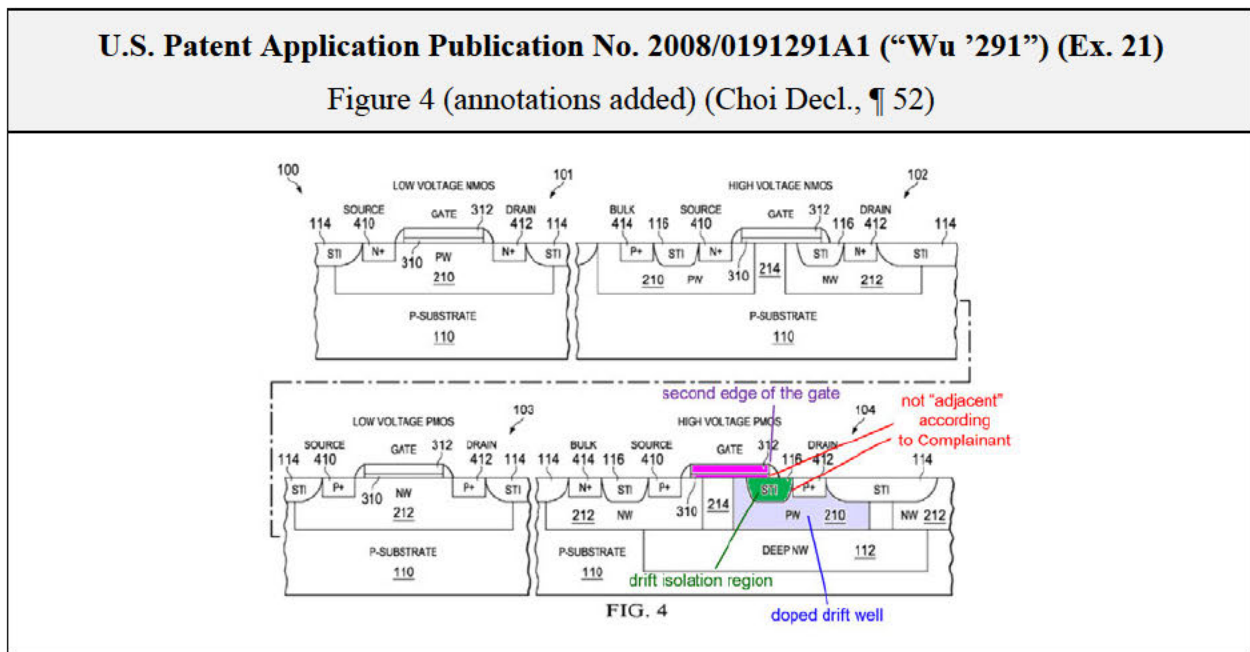


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depend on the gate voltage applications,” where “higher gate voltages may require thicker gate dielectric layers.” ’983 patent, 8:13-16. *See Icon Health & Fitness, Inc. v. Polar Electro Oy*, 656 F. App’x. 1008, 1016 (Fed. Cir. 2016) (affirming indefiniteness where “terms are a moving target that may change over time”).

The testimony of Tower’s expert David Choi confirms that a POSITA would not have had reasonable certainty as to “adjacent to.” Choi Decl., ¶¶ 41-53; *see also id.*, ¶¶ 28-37. Dr. Choi opines that a POSITA would have lacked any objective boundary for the scope of spatial relationships that are encompassed by the “adjacent to” term. Choi Decl., ¶¶ 41, 44-53. Dr. Choi opines that a POSITA would not have had reasonable certainty for the “adjacent to” claim language given the variety of differently situated elements the claims require to be “adjacent to” each other. *Id.*, ¶¶ 44-45. For example, a POSITA would have had no reasonable certainty for whether a distance of a picometer, a nanometer, a micrometer (microns), tens of microns, hundreds of microns, or longer is close enough to be considered “adjacent to” each other. *Id.*, ¶ 46. Dr. Choi further opines that a POSITA would have had no reasonable certainty for how many elements could come between two “adjacent” elements before they are no longer “adjacent to” each other. *Id.*, ¶ 47. Further, Dr. Choi opines that a POSITA would have had no objective boundary given the dual requirements that the “first heavily doped region” is “adjacent to” and “displaced from” the “first edge of the gate.” *Id.*, ¶¶ 46-50. Dr. Choi opines that the lack of reasonable certainty would have been especially acute in the context of laterally diffused metal oxide semiconductor (LDMOS) circuits, which have “significant variation in terms of the distances between laterally distributed structures in the device.” *Id.*, ¶ 47. Further, Dr. Choi opines that a POSITA’s lack of reasonable certainty would have been informed by the understanding that MOSFET transistors are three-dimensional devices. *Id.*, ¶ 46.

The lack of reasonable certainty for the scope of “adjacent to” is also apparent from Complainant’s attempts to distinguish the prior art by arbitrarily arguing that certain elements are not “adjacent to” each other. Choi Decl., ¶¶ 51-53. For example, Complainant argues regarding the prior art reference U.S. Patent Application Publication No. 2008/0191291A1 (“Wu ’291”) (Ex. 21) that “Tower has not established the asserted adjacency relationship between the drift well and the gate’s second edge from Wu ’291’s cited passages.” Ex. 32 at 66. The below demonstrative diagram shows the elements in the prior art that Complainant arbitrarily argues does not have the claimed “adjacency relationship”:



As another example, Complainant argues for the prior art reference U.S. Patent Application Publication No. 2009/0020811A1 (“Voldman ’811”) (Ex. 22) that “Tower has failed to explain how element 424 comprises the claimed doped drift well adjacent to a second edge of the gate.” Ex. 32 at 77. The below demonstrative diagram shows the elements Complainant is arguing without explanation are not “adjacent to” each other:

U.S. Patent Application Publication No. 2009/0020811A1 (“Voldman ’811”) (Ex. 22)

Figure 4 (annotations added) (Choi Decl., ¶ 53)

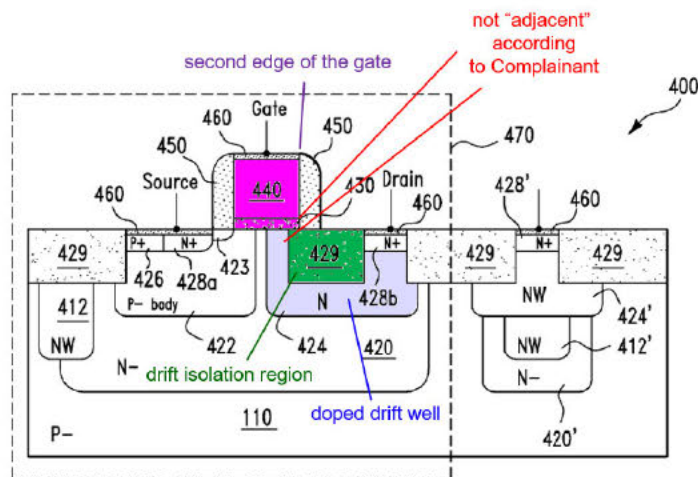


FIG. 4

A POSITA would have had no reasonable certainty for the “adjacent to” claim language, particularly given the unexplained, arbitrary limits on that scope Complainant appears to be asserting. Choi Decl., ¶¶ 51-53.

Since a POSITA would not have had reasonable certainty as to the scope of the “adjacent to” or “the channel is displaced from [a/the] first heavily doped region adjacent to the first edge of the gate” claim terms, the Court should find all asserted claims of the ’983 patent invalid as indefinite.

E. “the drift isolation region is disposed under the gate”

Disputed Claim Term	Complainant’s Proposed Construction	Respondents’ Proposed Construction
“the drift isolation region is disposed under the gate” (’983 patent, claims 4 and 16)	Plain and ordinary meaning	“the drift isolation region is positioned at least in part under the gate”

Tower’s construction seeks to clarify the plain-and-ordinary meaning of the claim language “the drift isolation region is disposed under the gate”—*i.e.*, that it covers any drift isolation region

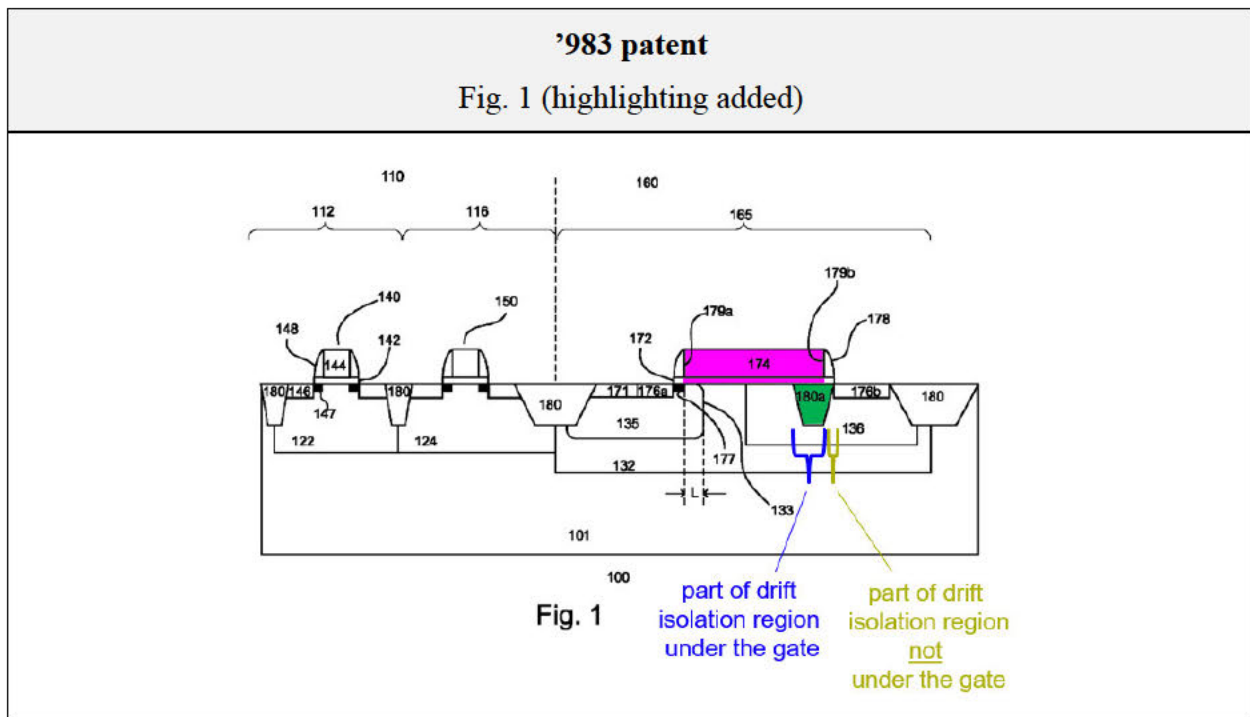
positioned at least in part under the gate. Complainant purportedly asserts that the claim language should be given its plain-and-ordinary meaning, but refuses to disclose how that differs from Respondents' proposed construction. Indeed, during parties' August 19 meet-and-confer call, Complainant indicated that it would likely be able to agree to Tower's proposed construction. But then, without explanation or describing any objection, Complainant announced that it did not agree with Tower's proposed construction. And Complainant's validity contentions suggest that it is relying on an unexplained and unsupported understanding of the "disposed under the gate" claim language contrary to the plain-and-ordinary meaning. For example, Complainant asserts for one prior art reference (which discloses a drift isolation positioned in part under and overlapping with a gate) that Tower has not shown "that the described overlap corresponds to the asserted under-the-gate arrangement." Ex. 32 at 86. Complainant should not be able to hide behind its unexplained plain-and-ordinary meaning proposal while at the same time arguing that the claims include some unspecified "under-the-gate arrangement" that is different from Tower's proposal. Accordingly, the Court should adopt Tower's construction, which clarifies the plain-and-ordinary meaning and avoids gamesmanship.

Claims 4 and 16 depend from claims 3 and 13, which recite "a drift isolation region in the doped drift well." '983 patent, 13:31-32, 14:17-18. According to claims 1 and 13, the doped drift well is "adjacent to a second edge of the gate." *Id.*, 13:27, 14:17. Claims 4 and 16 further recite that "the drift isolation region is disposed under the gate." *Id.*, 13:35-36, 14:27-28. Nothing in the claims suggests that any particular amount or portion of the "drift isolation region" needs to be "disposed under the gate." Rather, the claims recite simply that the drift isolation region—*i.e.*, at least part of the drift isolation region—be disposed under the gate.

The specification makes clear that the drift isolation region is disposed under the gate

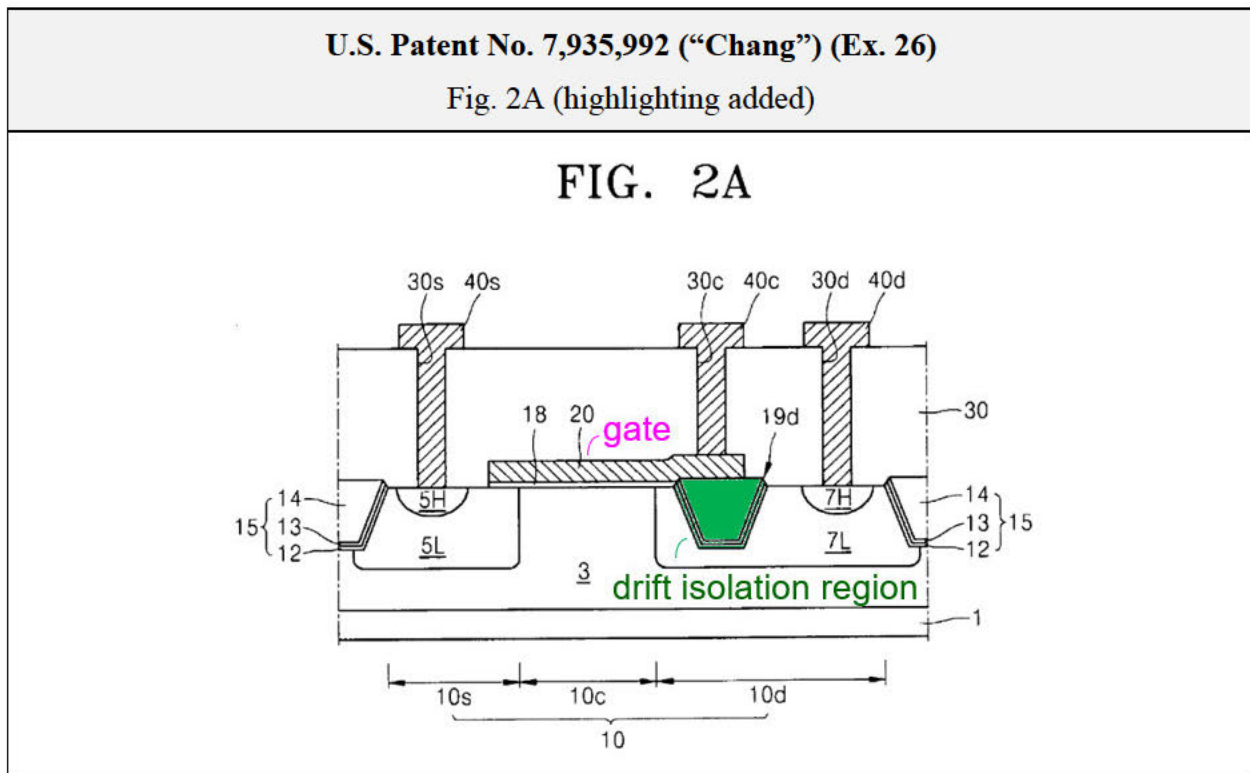
whenever it is positioned at least in part under the gate. For example, the specification discloses that “the drift isolation region **180a** is disposed in the drift well **136** under the gate” and the drift well can be in various positions, including “partially underlying the gate.” ’983 patent, 5:21-22. Since the drift well can be partially underlying the gate, the drift isolation region must also be able to be partially underlying the gate, since it is “disposed in the drift well[.]” *Id.*

The specification also discloses that a drift isolation region disposed “under the gate” is shown in Figure 1. ’983 patent, 5:21-22. Figure 1 shows drift isolation region **180a** situated partially under the gate—where the gate consists of “a gate electrode **144** over a gate dielectric **142.**” *Id.*, 3:30-31; *see also id.*, 13:17-19, 14:5-7, 14:43-45. The drift isolation region is only partially under the gate:



During prosecution, the examiner agreed (and the applicant did not contest) that a “drift isolation region ... disposed under the gate” can be a drift isolation region partially under the gate. Like asserted claims 4 and 16, original claim 5 of the ’983 patent required that the “drift isolation region is disposed under the gate.” Ex. 24 at GF-ITC-00004520. In an October 26, 2012 non-

final rejection, the examiner found that original claim 5 was obvious in light of the Negoro (U.S. Patent No. 6,979,864) (Ex. 25) and Chang (U.S. Patent No. 7,935,992) (Ex. 26) prior art. The examiner cited to Figure 2 of the Chang reference, reproduced below, which shows the “drift isolation region” extending beyond the gate:



Accordingly, the examiner recognized the plain-and-ordinary meaning of the language of asserted claims 4 and 16 reflected in Tower’s construction—*i.e.*, the claims are satisfied by a drift isolation region positioned at least in part under the gate. The applicant did not contest the examiner’s understanding, instead only arguing in an amendment that original claim 5 was not satisfied by the prior art following amendments to original claim 1 from which original claim 5 depended. Ex. 27 at GF-ITC-00004603; *see also Deere & Co. v. Bush Hog, LLC*, 703 F.3d 1349, 1358 (Fed. Cir. Dec. 4, 2012) (citing examiner statement and lack of applicant rebuttal).

Complainant’s apparent position that there is some arbitrary minimum amount of overlap between the gate and the drift isolation region necessary to satisfy claims 4 and 16 is an improper

narrowing of the claim scope that does not reflect the plain-and-ordinary meaning and has no support in the intrinsic or extrinsic evidence. The Court should adopt Tower’s construction to ensure that the plain-and-ordinary meaning of claims 4 and 16 is applied—*i.e.*, “the drift isolation region is positioned at least in part under the gate.”

F. “the channel having a channel length that is controlled by ion implantation”

Disputed Claim Term	Complainant’s Proposed Construction	Respondents’ Proposed Construction
“the channel having a channel length that is controlled by ion implantation” (’983 patent, claims 7 and 18)	Plain and ordinary meaning	“the channel having a channel length that is formed by implanting ions into the substrate to form a channel well”

In order to clarify the meaning of the claim, Tower has proposed a construction that adheres to the plain-and-ordinary language of the claim and the patent specification. While Complainant purports to also be giving the term its plain-and-ordinary meaning, Complainant’s validity contentions indicate it is taking an undisclosed and unsupported narrow construction in an effort to distinguish invalidating prior art. Complainant should not be permitted to hide behind plain-and-ordinary meaning, while at the same time attempting to distinguish the prior art based on some undisclosed and unsupported narrower understanding of the claim.

Tower’s invalidity contentions detail extensive prior art that teaches all limitations of the asserted claims, including ion implantation controlling channel length. In an effort to distinguish that prior art, Complainant has taken the position that the “controlled by ion implantation” claim language has some arbitrary, unexplained requirement not supported by the claim language or the evidence. For example, Complainant’s validity contentions argue that the “controlled by ion implantation” language requires the length of the channel to be “controlled by an implantation parameter,” which in Complainant’s view is “distinct from the formation of some doped region by

implantation[.]” Ex. 32 at 83-84. But nothing in the claims require “an implantation parameter.” And Complainant refuses to explain why the plain-and-ordinary meaning of a “channel having a channel length that is controlled by ion implantation” is “distinct from the formation of [the channel] by implantation.”

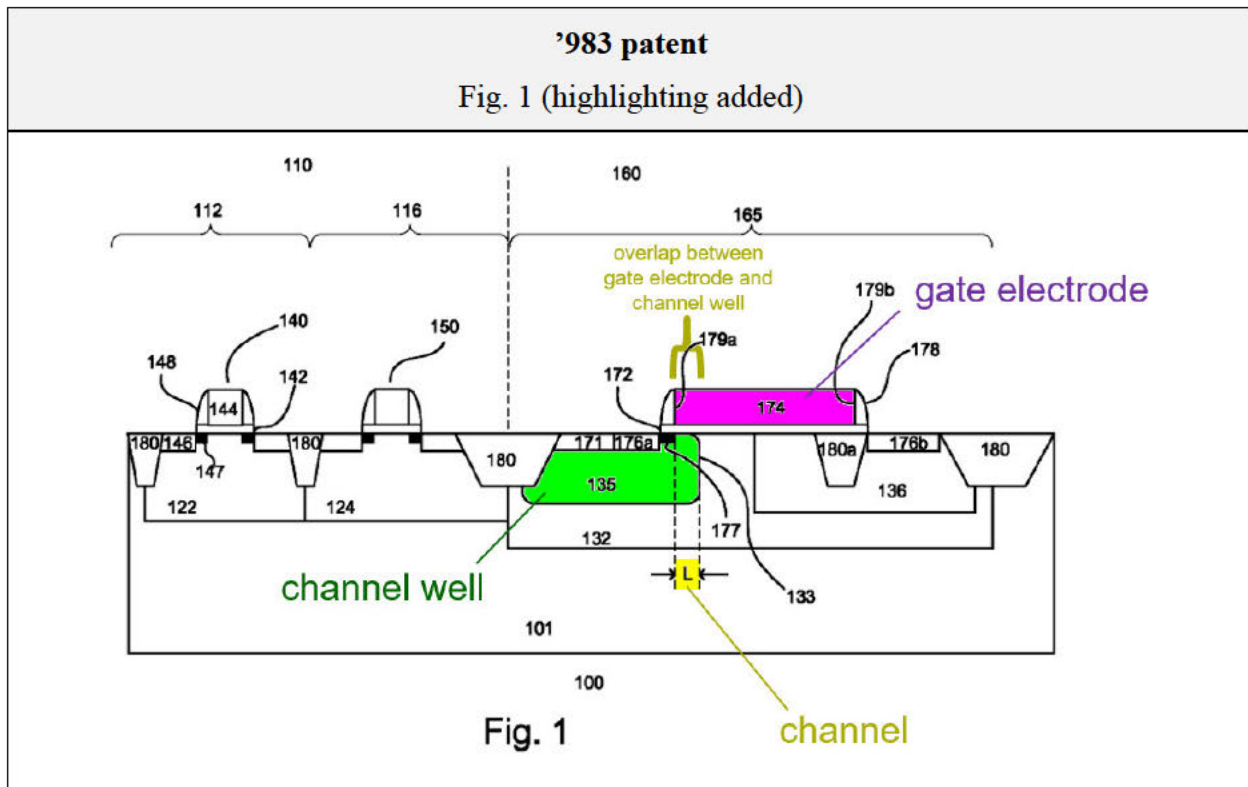
Unlike Complainant’s undisclosed and unsupported interpretation of the claims, Tower’s construction is supported by the language of the claims, the specification, and the file history. *First*, the language of claims 7 and 18 (which depend from claims 1 and 13, respectively) supports Tower’s position. Claims 1 and 13 both require a channel defined by the amount the “doped channel well” extends under the “gate,” such that the “first edge of the gate and channel edge beneath the gate define the channel of the device.” ’983 patent, 13:20-24, 14:8-14. A section of the substrate is doped to form a “doped channel well.” That doped channel well extends under the gate. The extent of that overlap between the doped channel well and the gate is the “channel.” Claims 7 and 18 further recite “a channel length that is controlled by ion implantation.” *Id.*, 13:49-50, 14:37-38. Since claims 1 and 13 make clear that the channel is defined by the overlap between the gate and the channel well, the plain meaning of claims 7 and 18 is simply that the channel is formed by implanting ions to form the channel well. The extent of the channel well determines the channel—so controlling the channel well through ion implantation means implanting ions to form the channel well.

Second, the specification confirms that the “controlled by ion implantation” claim language refers to nothing more than “implanting ions into the substrate to form a channel well.” *See, e.g.*, ’983 patent, 8:56-57 (“the substrate is implanted with dopants to form a channel well”), 8:60-62 (“the dopants are implanted at an angle to form a channel from the gate edge to the edge of the channel well”), 8:56-57 (“the substrate is implanted with dopants to form a channel doped well”),

1:34-38 (“forming short channel lengths” and “forming the channel well”), 7:37-41 (“The doped wells can be formed by ion implantation.”); *see also* ’983 patent, 10:30-32, 11:51-53, 12:36-38, 11:45-46, 12:30-31.

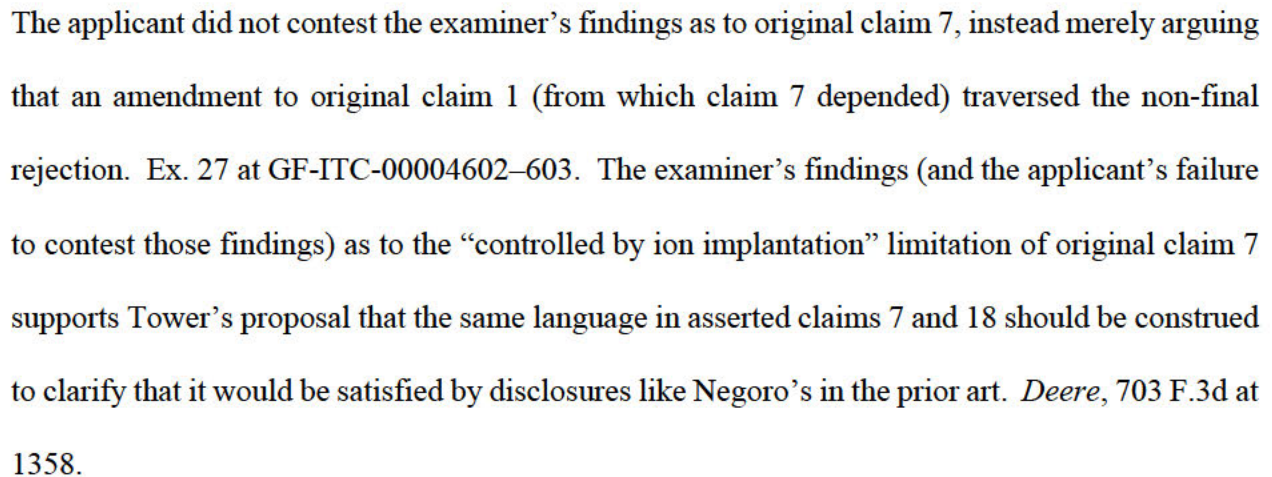
The only discussion of “control[]” of the channel length in the specification relates to formation, using dopant ions, of the channel well (which the specification also refers to as the channel or body well/implant). ’983 patent, 4:17-19 (“channel (or body) well **135**”). For example, the specification discloses that “[t]he effective channel length may be controlled by channel or body implant conditions selected to satisfy threshold voltage and breakdown voltage requirements.” *Id.*, 5:62-65. Further, the specification discloses using an implant mask to form a channel well, which “enables the channel length of the HV device to be well controlled to produce very short effective channel lengths[.]” *Id.*, 9:35-38. The specification’s discussion of ion implantation processes indicates that they result in the structure of the doped channel well, which determines channel length. *E.g.*, ’983 patent, 8:60-62 (“dopants are implanted at an angle to form a channel from the gate edge to the edge of the channel well”).

The specification also confirms that the channel is formed by the overlap of the channel well and the gate. For example, Figure 1 of the ’983 patent shows that the channel length “L” is defined by the amount the channel well **135** extends under the gate electrode **174**. *See Choi Decl.*, ¶¶ 28-37.



The specification discloses that “[t]he channel edge of the channel well 135 and the first edge 179 *a* of the gate on the source side define a channel of the HV transistor having an effective channel length *L*.” ’983 patent, 5:53-56.

Third, the file history of the ’983 patent further supports Tower’s proposed construction for claims 7 and 18. In an October 26, 2012 non-final rejection, the examiner for the ’983 patent application found that original claim 7 was satisfied by the Negoro prior art reference—U.S. Patent No. 6,979,864 (Ex. 25). Ex. 28 at GF-ITC-00004578–79. Similar to asserted claims 7 and 18, original claim 7 required that “the effective channel length is controlled by ion implantation.” Ex. 24 at GF-ITC-00004521. The examiner found that this limitation was satisfied by Negoro’s disclosure of the formation of a channel well. Ex. 28 at GF-ITC-00004579. Finding original claim 7 satisfied by Negoro, the examiner cited Negoro’s disclosure that “the surface of the P-type channel well region 23 under the N-type gate electrode 11g is operative as a channel region.” Ex. 25, 15:19-21 (cited by Ex. 28 at GF-ITC-00004579). The examiner further cited Negoro’s Figure



34

to control channel length. For example, the related U.S. Patent No. 8,053,319 claims “implanting ions...to form a channel well in the substrate...wherein a first portion of the channel well is between the first channel edge and the first mask opening edge, the first portion defines a channel with a length L of a device[.]” Ex. 29; *see also* Ex. 30 (“implanting ions...into the substrate to form a channel well”).

The intrinsic and extrinsic evidence makes clear that implanting dopant ions to form a channel well is sufficient to satisfy the claim language. No evidence supports Complainant’s undisclosed and unsupported position that claims 7 and 18 narrowly require channel length to be “controlled by an implantation parameter” and “distinct from the formation of some doped region by implantation.” Ex. 32 at 83-84. Accordingly, the Court should adopt Tower’s construction and foreclose Complainant from raising unsupported arguments.

G. “edge of the gate”

Disputed Claim Term	Complainant’s Proposed Construction	Respondents’ Proposed Construction
“edge of the gate” (’983 patent, claims 1, 2, 6, 13, 14, 15, and 20)	Plain and ordinary meaning	“edge of the gate electrode over the gate dielectric”

The term “edge of the gate” should be construed as “edge of the gate electrode over the gate dielectric” to clarify the plain-and-ordinary meaning. The claims make clear that the plain-and-ordinary meaning of the term “edge of the gate” is the “edge of the gate electrode over the gate dielectric”—*i.e.*, the edge of the region where there is a gate electrode over a gate dielectric. While Complainant has failed to articulate any objection to Tower’s construction or Complainant’s own understanding of the plain-and-ordinary meaning, Complainant’s remarks during the meet-and-confer process suggest that it may later argue that the “edge of the gate” could refer to an edge of the gate dielectric *without* any gate electrode over it. Since that interpretation is contrary to the

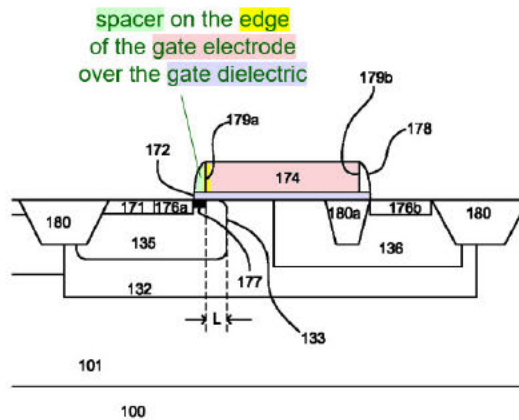
plain-and-ordinary meaning of “edge of the gate,” the Court should adopt Tower’s construction to clarify the plain-and-ordinary meaning and prevent Complainant from hiding an unsupported interpretation behind the mask of its unexplained “plain and ordinary meaning” proposal.

The claim language indicates that the “edge of the gate” refers to the edge of the gate electrode that is over the gate dielectric. The claims indicate that the “gate” includes “a gate electrode over a gate dielectric.” ’983 patent, 13:18-19, 14:6-7, 14:43-44. If there is no gate electrode, or if the gate electrode is not over the gate dielectric, then there is no “gate.” Accordingly, the “edge of the gate” is the edge of the region constituting the “gate electrode over a gate dielectric.”

Other claim language corroborates that the “edge of the gate” is the “edge of the gate electrode over the gate dielectric.” For example, claims 2 and 14 of the ’983 patent recite a “spacer on the first edge of the gate, wherein the spacer defines the displacement of the channel from the first heavily doped region.” ’983 patent, 13:28-30, 14:19-22 (emphasis added). Figure 1 shows where the claimed “spacer” needs to be situated to be “on the first edge of the gate” while defining the “displacement of the channel from the first heavily doped region.” The “spacer” is situated on the **edge of the gate electrode over the gate dielectric**:

'983 patent

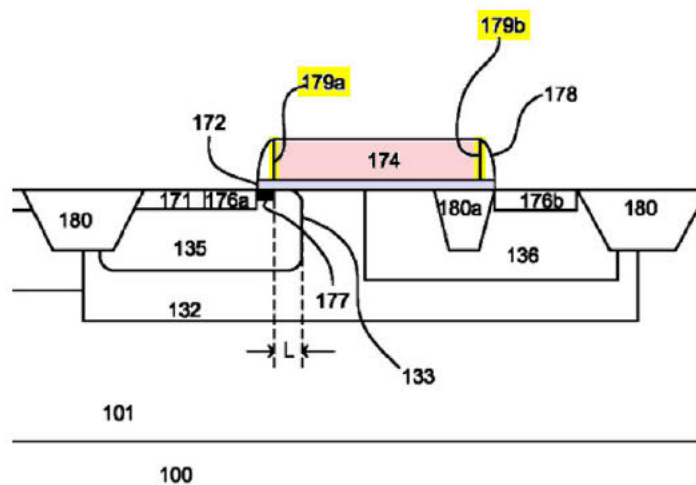
Fig. 1 (excerpt, highlighting added)



The specification confirms that the “edge of the gate” refers to the edge of the gate electrode over the gate dielectric. For example, in discussing Figure 1, the '983 patent specification discloses “a gate with first and second edges **179a-b** or sidewalls.” Figure 1 shows that the **first and second edges 179a-b** are the edges of the gate electrode over the gate dielectric:

'983 patent

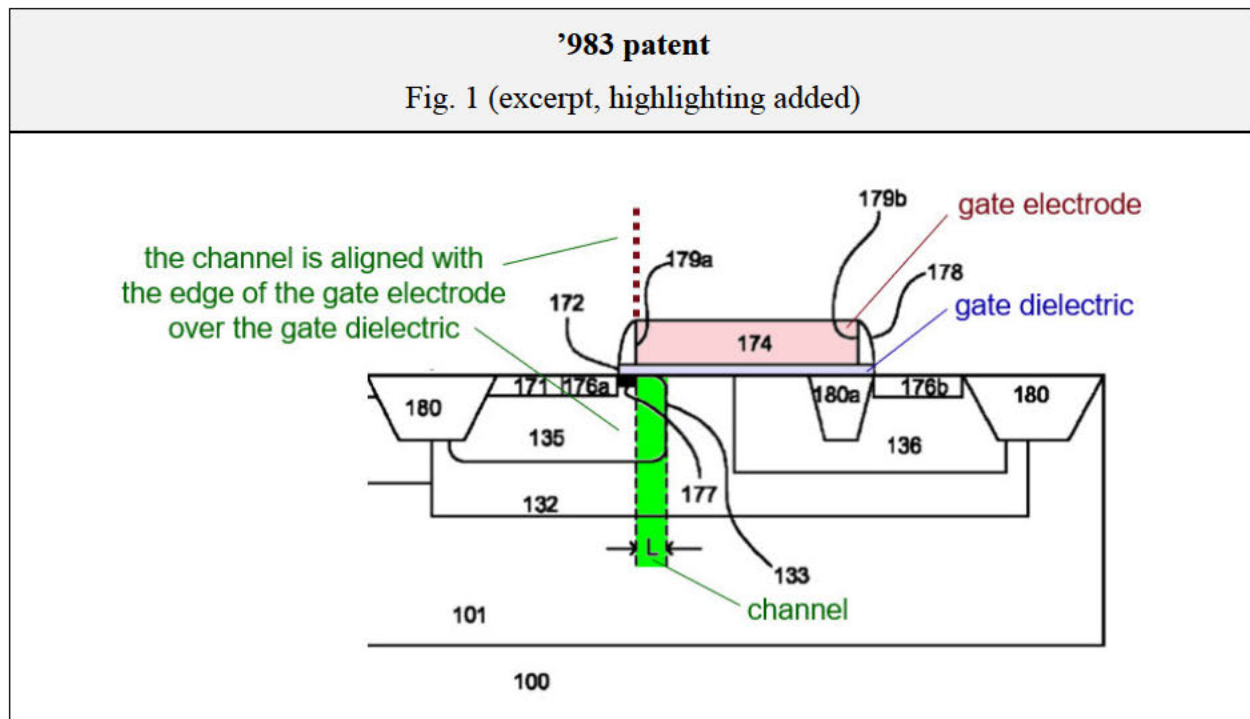
Fig. 1 (excerpt, highlighting added)



Further, the specification indicates that—based on the electrical characteristics of the

device—the “edge of the gate” must be the edge of the gate electrode over the gate dielectric. “A transistor...comprises a gate having a gate electrode 144 over a gate dielectric 142.” ’983 patent, 3:30-31. In other words, (1) a gate electrode over (2) a gate dielectric is necessary to have a “gate” that can function as a transistor for the claimed device of the ’983 patent.

The ’983 patent pegs the claimed “channel” to the edge of the gate electrode over the gate dielectric. All claims of the ’983 patent require that the “first edge of the gate and channel edge beneath the gate define a channel of the device.” ’983 patent, 13:23-24, 14:12-14, 14:51-53. Figure 1 shows the **channel** as the length “L” between two dotted lines. The leftmost edge of the channel—defined by the “first edge of the gate”—lines up with the edge of the gate electrode over the gate dielectric:



The extrinsic evidence confirms that a POSITA would have considered the claimed “edge of the gate” to be the edge of the gate electrode over the gate dielectric. Expert testimony from Tower’s expert Dr. David K. Choi opines that a POSITA would have understood that the gate of any metal oxide semiconductor field effect transistor (MOSFET) device such as the ’983 patent

discloses requires a gate electrode over a gate dielectric. Choi Decl., ¶¶ 54-56. Dr. Choi opines that a POSITA would have understood that such a device operates based on the gate electrode creating a channel on the bottom edge of the gate dielectric via an electric field created by the gate electrode when a voltage is applied to it. *Id.* Accordingly, a POSITA would have understood that the claimed “channel” must be pegged to the “edge of the gate over the gate dielectric.” *Id.*, ¶ 55. A POSITA would have understood that a channel could not be defined by an edge of a gate dielectric without a gate electrode over it, since there would be no electric field to form the channel. *Id.* Further, a POSITA would have understood that it would be nonsensical to consider an extended edge of a gate dielectric with no gate electrode over it to be the “first edge of the gate,” because a “gate” in a MOSFET is defined by its ability to close or open a channel using an electric field. *Id.*, ¶¶ 55-56.

Since the evidence indicates that the plain-and-ordinary meaning of “edge of the gate” is “edge of the gate electrode over the gate dielectric,” the Court should adopt Tower’s proposed construction.

IV. CONCLUSION

For the reasons above, there is clear and convincing evidence that all claims of the ’546, ’167, ’235, and ’983 patents are invalid for indefiniteness. However, if the CALJ is not inclined to find indefiniteness on this record, Respondents respectfully request that the CALJ defer consideration of indefiniteness until after the parties have developed the record during fact and expert discovery. In addition and/or the alternative, Respondents respectfully request adoption of their proposed constructions for the reasons above.

CERTIFICATE OF WORD COUNT

Pursuant to Ground Rules 1.7 and 7.2, the undersigned hereby certifies that the foregoing contains 9,993 words, including footnotes but excluding words in the caption, tables of contents/authorities/exhibits, figures, signature block, this Certificate of Word Count, and the Certificate of Service. Furthermore, this brief complies with the 10,000-word limit for opening claim construction briefs in Ground Rule 7.2. In preparing this certificate, the undersigned relied on the word count of the Microsoft Word program.

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CERTIFICATE OF SERVICE

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